# PG No. 25 $C_{6v}$ $6mm$ [ hexagonal ]

* generator : $3^{+}_{001}$, $2_{001}$, $\mathrm{m}_{110}$
* conjugacy class

$[1]$ : 1
$[2_{001}]$ : $2_{001}$
$[3^{+}_{001}]$ : $3^{+}_{001}$, $3^{-}_{001}$
$[6^{+}_{001}]$ : $6^{+}_{001}$, $6^{-}_{001}$
$[\mathrm{m}_{100}]$ : $\mathrm{m}_{100}$, $\mathrm{m}_{010}$, $\mathrm{m}_{110}$
$[\mathrm{m}_{120}]$ : $\mathrm{m}_{120}$, $\mathrm{m}_{210}$, $\mathrm{m}_{1-10}$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

| No. | tag | matrix (polar) | det |
|---|---|---|---|
| 1 | 1 | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 2 | $3^{+}_{001}$ | $\begin{bmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 3 | $3^{-}_{001}$ | $\begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 4 | $2_{001}$ | $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 5 | $6^{-}_{001}$ | $\begin{bmatrix} 0 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 6 | $6^{+}_{001}$ | $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | 1 |
| 7 | $\mathrm{m}_{110}$ | $\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | −1 |
| 8 | $\mathrm{m}_{100}$ | $\begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | −1 |
| 9 | $\mathrm{m}_{010}$ | $\begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | −1 |
| 10 | $\mathrm{m}_{1-10}$ | $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | −1 |
| 11 | $\mathrm{m}_{120}$ | $\begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | −1 |

*continued ...*

Table 1

| No. | tag | matrix (polar) | det |
|---|---|---|---|
| 12 | $m_{210}$ | $\begin{bmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | $-1$ |